

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 09:29 UTC

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Generated 2026-05-17 09:29 UTC

- Cycles recorded: **738**
- Time span: 563.8h (~1.31 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	700
BARRIER_HIT	19
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-17 02:29 UTC	INCONCLUSIVE	Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm
2026-05-17 02:59 UTC	INCONCLUSIVE	Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP
2026-05-17 03:06 UTC	BARRIER_HIT	Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC
2026-05-17 03:32 UTC	INCONCLUSIVE	q-Major Cancellation Gap Separates Permanent from Determinant Sup
2026-05-17 04:06 UTC	INCONCLUSIVE	Immanant Positivity Width Bounds Monotone KW Depth for Perfect Ma
2026-05-17 04:32 UTC	INCONCLUSIVE	Curto-Fialkow Hankel Defect Separates Read-Twice IP_2 BPs
2026-05-17 05:07 UTC	INCONCLUSIVE	Hodge-Cheeger Product Lower-Bounds Tseitin Resolution Length
2026-05-17 05:45 UTC	INCONCLUSIVE	Sandpile Group Order Lower-Bounds Monotone KW Depth
2026-05-17 06:17 UTC	INCONCLUSIVE	Conway Temperature Bounds Tree-Resolution Width for 3-SAT
2026-05-17 06:44 UTC	INCONCLUSIVE	Hochster Regularity of Variable Co-Occurrence Graph Lower-Bounds
2026-05-17 07:23 UTC	INCONCLUSIVE	Ihara Zeta Entropy Predicts Tseitin DPLL Tree Size

Time	Verdict	Title
2026-05-17 08:06 UTC	INCONCLUSIVE	Signed-Support L_{inf} Discrepancy Bounds Bottom Fan-In for AC0 PAR
2026-05-17 08:50 UTC	INCONCLUSIVE	2-Sylow Rank of Critical Group Bounds Tseitin DPLL Size
2026-05-17 09:02 UTC	INCONCLUSIVE	Stable-Rank of Layer-Symbol Matrix Separates Read-Twice BPs for I
2026-05-17 09:29 UTC	INCONCLUSIVE	TP_2 Minor-Sign Defect of Spectral Pseudo-Moment Matrix Bounds Ma

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```